Intelligent Multi-Agent Rainwater Management for Urban Flood Mitigation

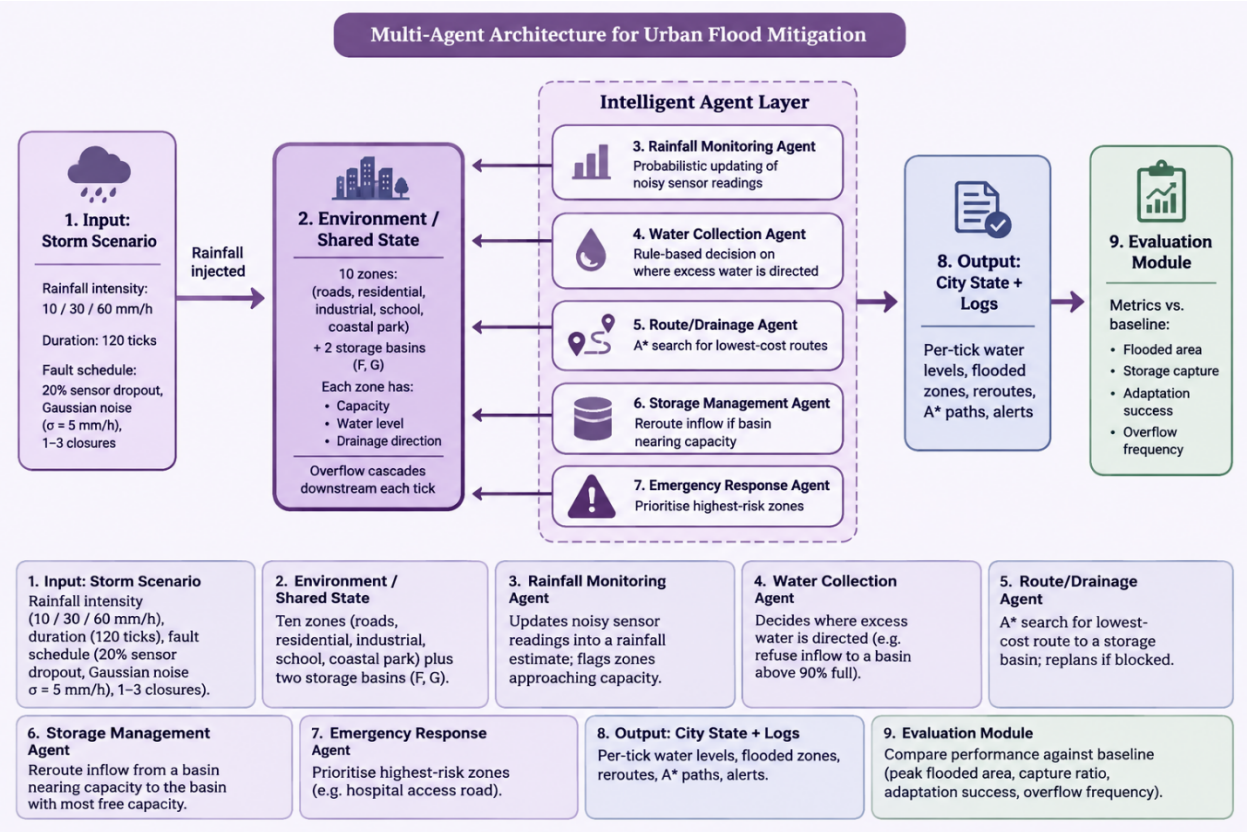
CS3081 – Report 3: Proposed AI Solution & Architecture Diagram

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1. Proposed Solution

The problem, as stated in Report 1, is to decide each minute how to route inflow among ten zones and two storage basins to minimise flooded zones under uncertain rainfall and closed drainage connections. We address this with a multi-agent testbed in which five agents — Rainfall Monitoring, Water Collection, Route/Drainage, Storage Management and Emergency Response — cooperate through one shared environment model, directly targeting the Report 2 gap that no reviewed system combines noisy-sensor reasoning, hazard-aware routing and basin allocation during the event itself.

2. Architecture Diagram



3. Main Components

The environment model is the single source of truth all five agents read and write. Rainfall Monitoring converts noisy readings into a probabilistic rainfall estimate. Water Collection decides by rule whether a zone's excess water is admitted to a basin. Route/Drainage supplies A* paths around blocked connections and replans. Storage Management resolves contested basin capacity by threshold rule. Emergency Response applies a fixed priority rule base when water cannot be moved in time. The evaluation module aggregates metrics per run against the baseline.

4. AI Methodology

- **Probabilistic updating (Rainfall Monitoring).** Each zone holds a prior rainfall-intensity estimate, revised each step by a noisy sensor likelihood, so a dropped or conflicting reading shifts the estimate instead of being trusted outright — targeting Report 2's "sensing without action" gap, which no reviewed paper closes for stormwater agents [10]–[12].
- **A* search** (Route/Drainage). State: a zone. Actions: move to a connected zone along an open drainage edge. Goal: reach a storage basin. Heuristic: a graph-distance estimate to the nearest basin (e.g. number of drainage edges weighted by capacity cost); admissible because true drainage cost is never shorter than the edge-count estimate. Constraint handling: closed connections are removed from the search space and replanned around, rather than treated as fixed obstacles as in [6], [7] — the "obstacles, not hazards" gap from Report 2. Justified because the graph is small and A* with an admissible heuristic returns shortest safe paths, matching the hazard-aware routing evidence in [9]. A* with an admissible heuristic expands fewer nodes than Dijkstra on this graph while returning the same optimal path, and BFS is unsuitable because edge costs are non-uniform, so BFS would not return a minimum-cost route**.**
- **Rule-based reasoning (Water Collection, Storage Management, Emergency Response).** Deterministic threshold and priority rules (basin above 90% full: refuse inflow; fill ratio ≥ threshold: reroute to the basin with most free capacity; overflow: rank by criticality). Justified because these are operating policies, and rule-based knowledge representation is an approved classical technique.

No machine learning is used.

5. Evaluation Strategy

Test cases follow Report 1: light, heavy and extreme rainfall (10 / 30 / 60 mm/h) over 120 steps, with 1–3 random drainage-connection closures and, separately, 20% sensor dropout with Gaussian noise ($\sigma = 5$ mm/h), 30 runs per scenario.

Baseline: fixed drainage directions computed once, no replanning.

Measures: mean peak number of flooded zones vs. baseline; fraction of generated runoff captured; replanning success within 5 steps; storage utilisation; and the probabilistic estimate's mean absolute error against raw-sensor error; fraction of runs in which the highest-criticality zone (hospital access road) is not the first to overflow.

The Storage Management agent's threshold-based rerouting is already implemented and tested, including a stranded-zone edge case, giving early evidence that the design is implementable rather than only theoretical.

Appendix: AI Use Declaration

(a) **Evaluation Table**

Criterion	Score	Strengths	Weaknesses	Recommended Fix
Alignment with problem and research gap	1.5 / 2	Directly targets the central Report 2 gap: no reviewed system combines Bayesian sensing under noise, hazard-	The project explicitly leaves one of the five gaps open (gap 3: route	Either (a) add a simple risk-cost term to the A* edge weight (e.g., penalise edges adjacent to high-water

		aware routing, and resource allocation during the storm itself, coordinated by multiple agents. The five agents map cleanly onto the four open gaps identified (sensing-without-action, action-without-uncertainty, routing-without-storage). The Jeddah motivation is concrete and cited.	weighting by flood risk — closures treated as binary, not cost-weighted hazards). This is acknowledged honestly, but it means the system only partially closes the "obstacles, not hazards" gap that Report 2 identifies as the core weakness of A* planners [6]–[9]. The claim to address "four of these five gaps" is slightly generous: gap 4 (routing-storage coupling) is only addressed by threshold-based rerouting, which Report 2 itself calls "a first step," not a coupling.	zones) to close gap 3, or (b) explicitly reframe the contribution as "three gaps addressed, two partially" and justify why full hazard-weighted routing is out of scope for a 120-step simulation.
Architecture clarity and data flow	2.5 / 3	The architecture diagram is coherent: input → shared environment → five-agent layer → output/logs → evaluation. The shared environment model as single source of truth is well-motivated and matches the blackboard coordination in [13]. The data flow from noisy sensors → probabilistic estimate → downstream agents is explicit. The mapping of each agent to a technique is clear in Report 3, Section 4.	Unclear data flow between Rainfall Monitoring and the other agents. The diagram shows Rainfall Monitoring feeding the environment state, but it is not specified how the posterior rainfall estimate is consumed by Water Collection, Route/Drainage, or Storage Management. Does Water Collection use the posterior to decide admission? Does Route/Drainage use it to weight edges? If not, the "sensing without action" gap	Add one sentence per consumer agent stating exactly which fields of the shared state it reads (e.g., "Water Collection reads posterior_rainfall[zone] and basin_fill_ratio"). Define the Emergency Response agent's trigger condition precisely (e.g., "activates only when a zone exceeds 90% capacity and no admissible route exists").

			is only closed at the state level, not at the decision level. Emergency Response overlaps with Water Collection and Storage Management: all three are rule-based, and the boundary between "where excess water is directed" (Water Collection) and "which zones are prioritised" (Emergency Response) is fuzzy. Evaluation module is outside the agent loop — fine, but it should be explicit that it observes logs, not the live state.	
AI methodology and technical justification	1.5 / 2	Techniques are correctly classical: probabilistic updating (Bayesian), A* with admissible heuristic, and rule-based reasoning. Key elements are defined: A* state/actions/goal/heuristic; prior/likelihood/update for rainfall; rule conditions for thresholds. The justification of A* over Dijkstra (fewer expansions, same optimal path) and BFS (non-uniform costs) is correct and well-argued. The graph-distance heuristic weighted by capacity cost is a sensible fix over Euclidean distance. No ML is used, as required.	The Bayesian update is under-specified. "Prior rainfall-intensity estimate, revised each step by a noisy sensor likelihood" — what is the likelihood model? Gaussian with σ = 5 mm/h? What is the prior (previous step's posterior, or a climatological prior)? Is the update conjugate (Gaussian-Gaussian) or a particle filter? Without this, "probabilistic updating" is a label, not a method. The A heuristic admissibility claim needs care:* "graph-	Specify the Bayesian model formally (prior, likelihood, posterior, update equation) and state whether it is conjugate. Prove or bound the A* heuristic (e.g., "edge weight = 1 + capacity_cost, heuristic = min edge weight × graph distance, hence admissible"). Justify rules as encoding real operating policies (e.g., "90% threshold mirrors Jeddah municipality's basin operating rule" — or state it is a design choice).

			distance weighted by capacity cost" — if the weight is capacity cost, the heuristic may overestimate true drainage cost unless the weighting is a lower bound. The report asserts admissibility but does not prove it. Rule-based agents are not justified against alternatives — why thresholds rather than a simple priority queue or a greedy allocator? The "approved classical technique" justification is weak; it should argue why rules are appropriate (interpretability, operating-policy fidelity), not just that they are allowed.	
Feasibility of implementation	1.5 / 2	The system is small (10 zones, 2 basins, 120 steps), which is realistic for a course project. Storage Management is already implemented and tested, including a stranded-zone edge case — good early evidence. The choice of A* over heavier planners is appropriate for a small graph. The evaluation runs (30 per scenario × 3 rainfall rates = 90 runs, plus fault injection) are computationally trivial.	Scope creep risk: five agents, Bayesian updating, A* replanning, fault injection, and a full evaluation matrix is ambitious for the remaining timeline if only Storage Management is implemented. The report does not state what is not implemented (Route/Drainage is "in development" — how far?). The synthetic Jeddah-like district is not grounded: where do the ten zones,	Add a short "Implementation status" table with columns for Agent, Status, and Tested? State the simulation platform and the source of the zone graph (hand-authored is fine, but say so). If time is short, cut Emergency Response to a stub and document it as future work rather than half-implementing five agents.

			capacities, and drainage directions come from? Are they hand-authored? If so, the evaluation is on a toy graph, which weakens external validity. No mention of the simulation engine (Python? NetLogo? custom?) or how time steps are synchronised across agents.	
Evaluation strategy and metrics	0.5 / 1	Baseline (fixed drainage, no replanning) is defined and appropriate. Scenarios (10/30/60 mm/h), run counts (30 per scenario), fault parameters (20% dropout, σ = 5 mm/h, 1–3 closures), and the 5-step replan bound are all specified. Metrics map to objectives: flooded zones, runoff captured, replan success, storage utilisation, MAE of rainfall estimate, and critical-zone overflow. The inclusion of a rainfall-estimate MAE metric is a good, gap-specific measure.	Objective 1's "30% reduction in mean peak flooded area" is not reported in Report 3's evaluation plan — the plan lists "mean peak number of flooded zones vs. baseline" but does not commit to the 30% threshold. Objective 3's "45 of 50 runs" is inconsistent with the stated 30 runs per scenario; the report mixes 30-run and 50-run designs. No statistical test is proposed (e.g., paired t-test or bootstrap CI) for the 30-run comparisons, so "beats baseline" may be noise. Replanning success within 5 steps is measured, but not replanning quality (does the new route avoid flooding, or just avoid closed edges?). The 60 mm/h extreme	Reconcile run counts (30 or 50 — pick one and use it everywhere). Add a paired statistical test and report effect sizes with confidence intervals. Define a "replan quality" metric (e.g., post-replan flooded-zone count vs. pre-fault). Add at least one ablation (no Bayesian update, no replanning) to attribute gains. Check that the 60 mm/h scenario is discriminative (baseline should not saturate).

	scenario may flood regardless of agent behaviour — if the baseline floods all zones and the MAS floods 70%, the 30% target is met trivially; if both flood all zones, the target is unmeasurable. No ablations (e.g., MAS without Bayesian updating) to isolate which component drives improvement.

Total: 7.5 / 10

Are the selected AI techniques appropriate and correctly classical?

Yes, broadly. Probabilistic updating, A* search, and rule-based reasoning are all classical AI techniques from the course, and no ML/DL is used. The key elements are named — A* state/actions/goal/heuristic, Bayesian prior/likelihood/update, rule conditions — which is more than most student reports do.

However, the definitions are incomplete in two places. The Bayesian update lacks a likelihood model and prior specification, so it is a label rather than a method; and the A* heuristic's admissibility is asserted, not shown. The rule-based agents are justified only by "approved classical technique," which is a permission argument, not a design argument. These are fixable with two paragraphs.

2. Is the architecture logically connected and technically correct? Unnecessary modules or unclear flows?

Logically connected, yes. The shared environment as single source of truth is correct and matches the blackboard pattern in [13]. The input → state → agents → logs → evaluation flow is sound.

Two unclear flows and one overlap:

- **Rainfall Monitoring → downstream agents**: the posterior estimate is written to state, but it is not specified which agents read it or how it changes their decisions. If no agent consumes it, the "sensing without action" gap is not actually closed at the decision level.
- **Emergency Response vs. Water Collection/Storage Management**: all three are rule-based, and the boundary is fuzzy. Emergency Response should have a distinct trigger (e.g., "activated only when a critical zone is at risk and no admissible route exists") or be merged into Water Collection.
- **No unnecessary modules** — all five agents map to a stated gap. The evaluation module is correctly outside the loop.

3. Does the solution address the identified research gap?

Partially, and honestly so. It closes gap 2 (sensing-without-action) at the state level and gap 5 (weak evaluation) via the baseline and fault injection. It makes a first step on gap 4 (routing-storage coupling) via threshold rerouting, but Report 2 itself calls this "a first step," not a coupling. Gap 3 (action-without-uncertainty / binary closures) is explicitly left open. The claim to address "four of five gaps" should be softened to "three addressed, two partially addressed," or gap 3 should be closed with a risk-weighted A* edge cost.

4. Are test cases, baseline, and evaluation measures realistic and sufficient?

Realistic in scale, insufficient in rigour. The 10-zone graph, 120-step horizon, and 30-run design are feasible. The baseline is appropriate. The fault parameters are well-chosen and match Objective 3.

Insufficient because: (a) the 30-run vs. 50-run inconsistency must be fixed; (b) no statistical test or CI is proposed, so "beats baseline" is unverifiable; (c) the 30% reduction target from Objective 1 is not carried into the evaluation plan; (d) no ablation isolates the Bayesian component's contribution; (e) the 60 mm/h scenario may saturate and fail to discriminate. These are one-page fixes, not redesigns.

Three Highest-Priority Improvements (realistic, one-page)

1. *Close or explicitly scope gap 3 with a risk-weighted A edge cost.** Add a simple flood-risk penalty to the edge weight (e.g., $w(e) = 1 + \alpha \cdot \max(0, \text{water_level}[to] - 0.8 \cdot \text{capacity}[to])$) and state the heuristic remains admissible because the penalty is non-negative. This turns "binary closures" into "hazard costs" and directly answers Report 2's core critique of [6]–[9]. If time is short, instead rewrite the gap claim as "three addressed, two partial" and defend the scope in one sentence.
2. *Formalise the Bayesian update and the A heuristic in half a page.** Write the prior (previous posterior, Gaussian), likelihood (Gaussian, $\sigma = 5$ mm/h), and conjugate posterior update equation; state the edge-weight definition and prove $h(n) \leq$ true cost (e.g., $h = \text{min_edge_weight} \times \text{graph_distance}$). This converts two labelled techniques into two specified methods and is the single largest score gain available in the methodology criterion.
3. **Reconcile the evaluation design and add one ablation.** Pick 30 runs (consistent with Report 1) or 50, not both; carry the 30% flooded-area target into the metrics table; add a paired test (e.g., Wilcoxon signed-rank) with effect size; and add one ablation ("MAS without Bayesian updating" or "MAS without replanning") to attribute gains. This is a table edit plus two sentences and directly addresses the weakest criterion (0.5/1).

(b) Changes made in response

1. Aligned Report 3 to Report 1's five-agent architecture: added the Water Collection and Emergency Response agents, which the earlier draft omitted, so the architecture report and the project description describe the same system.
2. Replaced the Euclidean-distance heuristic with a graph-distance heuristic weighted by capacity cost, because the ten-zone environment has no coordinate system and the original heuristic was not computable.
3. Unified terminology across Reports 1–3: "drainage-connection closures" replaces "road closures" and "route closures," matching the environment's drainage-edge model.

4. Specified each technique's key elements (state, actions, goal, heuristic; prior, observation, update; rule conditions) rather than naming techniques alone, and justified each choice against a source in Report 2 ([9] for hazard-aware A*; [10]–[12] for probabilistic sensing).
5. Stated explicitly that the Storage Management rerouting is implemented and tested while Route/Drainage is in development, so the evaluation claim is accurate and not overstated.

(c) AI tools used

I used Claude (Anthropic) to draft and revise the proposed solution, architecture description and evaluation strategy. I specified the system design and techniques myself, checked that the architecture matches the implemented code and the objectives in Report 1, and rewrote the methodology section after verifying the technique descriptions against the course material and the reviewed papers.